

Numpy

```
In [2]: import numpy as np
```

```
In [5]: import time
```

```
In [9]: size = 1000000000
```

```
In [ ]: py_list = list(range(size))
start = time.time()
result = [x*2 for x in py_list]    # list comprehension
print(f"Python list time: {time.time() - start:.5f}s")
```

```
In [8]: np_arr = np.arange(size)
start = time.time()
result = np_arr*2
print(f"Numpy list time: {time.time() - start:.5f}s")
```

Numpy list time: 0.00397s

Creating Arrays

```
In [3]: a = np.array([1,2,3,4,5])
```

```
In [4]: a
```

```
Out[4]: array([1, 2, 3, 4, 5])
```

```
In [5]: type(a)
```

```
Out[5]: numpy.ndarray
```

```
In [11]: import numpy as np
```

```
In [16]: a = np.array([1,2,3,4,5])    #1d array
```

```
In [17]: a
```

```
Out[17]: array([1, 2, 3, 4, 5])
```

```
In [18]: type(a)
```

```
Out[18]: numpy.ndarray
```

```
In [20]: a2 = np.array([[1,2,3],[4,5,6],[7,8,9]])
```

```
In [21]: a2
```

```
Out[21]: array([[1, 2, 3],
               [4, 5, 6],
               [7, 8, 9]])
```

```
In [22]: a3 = np.array([[[1,2],[3,4]], [[5,6],[7,8]], [[9,1],[3,4]]])
```

```
In [23]: a3
```

```
Out[23]: array([[1, 2],
               [3, 4]],

              [[5, 6],
               [7, 8]],

              [[9, 1],
               [3, 4]])
```

```
In [50]: int_arr = np.array([1,2,3], dtype=np.int32)
float_arr = np.array([1,2,3], dtype=np.float64)
bool_arr = np.array([0,1,1,1,0,0,1], dtype=bool)
comp_arr = np.array([1, 3], dtype=complex)
```

```
In [51]: print(f"int_arr : {int_arr.dtype}")
print(f"float_arr : {float_arr.dtype}")
```

```
int_arr : int32
float_arr : float64
```

```
In [52]: comp_arr
```

```
Out[52]: array([1.+0.j, 3.+0.j])
```

```
In [49]: print(f"comp_arr : {comp_arr.dtype}")
```

```
comp_arr : complex128
```

```
In [57]: t_arr = np.array( ((1,2,3),(4,5,6)) )
```

```
In [58]: t_arr
```

```
Out[58]: array([[1, 2, 3],
               [4, 5, 6]])
```

```
In [59]: type(t_arr)
```

```
Out[59]: numpy.ndarray
```

```
In [60]: t_arr.ndim
```

```
Out[60]: 2
```

```
In [63]: a3.ndim
```

```
Out[63]: 3
```

```
In [ ]: a = np.array([1,2,3,4,5,6,7,8,9], )
```